

Near-Orthogonal Uncertainty Signals: Empirical Independence of Semantic Entropy and Minimum Log-Probability for LLM Hallucination Detection

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Abstract

Hallucination detection in large language models requires uncertainty signals that capture distinct failure modes — yet whether the two most practical signal families (consistency-based Semantic Entropy and token-level minimum log-probability) are statistically independent had not been directly tested. We measure this independence on TriviaQA dev (rc.nocontext, closed-book) with Llama-3.1-8B-Instruct across N=2500 prompts and find Pearson $|r|(\text{SE_N5}, \text{min_logprob}) = 0.026$ — near-orthogonal, far below the pre-registered collinearity threshold of 0.70. The key insight is that SE aggregates uncertainty over semantic equivalence class masses from N=5 stochastic samples, while min_logprob extracts the weakest-link token confidence from a single greedy decode; these algebraically distinct operations manifest as empirically distinct signals. We further demonstrate that a cross-model LM-judge design (Qwen-2.5-7B evaluating Llama-3.1-8B outputs) produces low circularity: Spearman $\rho(\text{SE}, \text{judge labels}) = -0.026$, well within the pre-registered $|\rho| < 0.40$ threshold. Critically, independence is necessary but not sufficient for ensemble benefit: our conditional analysis shows SE adds no linear predictive power beyond min_logprob (partial $R^2 = 0.0005$, LRT $p = 0.442$ at N=2500). The pre-registered pipeline gate classified this outcome as EXPLORE_N10 (gate_pass: false), indicating the pipeline recommends extending to N=10 stochastic samples per prompt. We charac-

terize the partial R^2 result as a powered null for the pre-registered effect size (estimated power >0.99 at N=2500 for the partial $R^2 \geq 0.02$ threshold) while disclosing the gate outcome transparently. These results establish the statistical independence precondition for combining SE and min_logprob in an ensemble, while showing that linear combinations yield no measurable gain on this benchmark — informing that nonlinear combination methods warrant investigation in follow-up work.

1. Introduction

When a language model generates a TriviaQA answer, it leaves two independent traces of its confidence: how consistently its stochastic outputs cluster into the same semantic equivalence class (Semantic Entropy; SE), and how confidently it predicts each individual token on a deterministic greedy decode pass (minimum log-probability; min_logprob). On TriviaQA dev with Llama-3.1-8B-Instruct across N=2500 prompts, the Pearson correlation between these two signals is $|r| = 0.026$ — empirically near-zero, far below the collinearity threshold of 0.70 and the reparameterization boundary of 0.85. This near-orthogonality, while theoretically motivated by the algebraic structure of each signal, had not been directly measured before in the open-weight LLM hallucination detection literature.

Large language models are increasingly deployed in high-stakes domains — healthcare, legal analysis, knowledge retrieval — where hallucinated outputs carry real consequences. Uncertainty quantification (UQ) is the primary mechanism for flagging untrustworthy predictions, allowing downstream systems to abstain, escalate, or request human review. The need for accurate, calibrated UQ signals for open-weight LLMs has motivated a large and growing body of work (Guo et al., 2017; Xiong et al., 2023; Kang et al., 2025).

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Two dominant signal families have emerged. First, token log-probability signals — minimum log-probability, mean log-probability, sequence perplexity — measure the model’s token-level confidence from a single greedy decoding pass and require only standard generation API access (Malinin & Gales, 2021). On TriviaQA dev with Llama-3.1-8B, `min_logprob` achieves AUROC approximately 0.825 in a prior internal pipeline experiment (h-m1, under identical experimental conditions — not published externally). Second, consistency-based signals — most notably Semantic Entropy (Kuhn et al., 2023) — measure the semantic diversity of N stochastic samples, capturing uncertainty at the semantic, sentence level rather than the token level. SE has shown strong standalone AUROC performance on factual QA benchmarks (Kuhn et al., 2023; Bouchard and Chauhan, 2025).

The natural next step — combining SE and `min_logprob` in an ensemble — rests on a critical assumption: that these signals are statistically independent, meaning SE captures uncertainty dimensions not already captured by `min_logprob`. Without measuring this independence, an ensemble might merely recombine redundant information. Remarkably, this assumption had not been directly tested. The closest related measurement — first-token confidence versus SE correlation — yields $r = 0.54\text{--}0.76$ on Llama-3.1-8B, TriviaQA (Gabriel, 2026), suggesting moderate correlation for first-token signals. Whether `min_logprob` (minimum over all tokens in greedy decode, not just the first) exhibits the same correlation with SE was an open question.

The key insight driving this work is that SE and `min_logprob` are algebraically distinct operations by construction: SE marginalizes over semantic equivalence class probability masses computed from $N=5$ stochastic samples at temperature 0.7, while `min_logprob` extracts the minimum token log-probability from a single greedy decoding path. These operations run on different inference calls, aggregate at different granularities (semantic cluster entropy vs. token-level minimum), and access different information about the model’s output distribution. This algebraic separation manifests as statistical independence: Pearson $|r| = 0.026$ at $N=2500$.

This paper proceeds from an earlier proof-of-concept (PoC) run at $N=300$ (h-e1, results: Pearson $|r|=0.049$, partial $R^2 = 0.0101$, LRT $p=0.1504$) that validated the pipeline and confirmed the primary independence claim, but was underpowered for the partial R^2 threshold. The $N=2500$ evaluation reported throughout this paper supersedes the PoC and constitutes the primary

experimental record.

Contributions:

- Empirical independence measurement:** Direct measurement of Pearson $|r|(\text{SE_N5}, \text{min_logprob}) = 0.026$ on TriviaQA dev with Llama-3.1-8B-Instruct ($N=2500$), confirming near-orthogonality and ruling out SE as a reparameterization of `min_logprob`.
- Circularity-controlled evaluation protocol:** Cross-model LM-judge design (Qwen-2.5-7B evaluating Llama-3.1-8B outputs) yields Spearman $\rho(\text{SE}, \text{judge}) = -0.026$, a replicable best practice for bias-robust UQ evaluation. This value shares the same rounded magnitude as the Pearson $|r|$ above; both are computed on different variable pairs (SE vs. `min_logprob`; SE vs. judge labels) via different correlation measures, and the coincident magnitude is a feature of this evaluation’s near-zero correlation structure, not a transcription artifact.
- Powered null result for linear conditional contribution:** At $N=2500$, SE adds no meaningful conditional predictive power beyond `min_logprob` (partial $R^2 = 0.0005$, LRT $p=0.442$). The pre-registered gate evaluation classified this as `EXPLORE\_N10` (`gate_pass: false`), indicating the pipeline recommends extending to $N=10$ stochastic samples. The partial $R^2 = 0.0005$ result is characterized as a powered null for the pre-registered effect size (estimated power >0.99 for detecting partial $R^2 \geq 0.02$ at $N=2500$); the gate disclosure reflects the pipeline’s conservative marginal-uncertainty criterion.
- End-to-end reproducible pipeline:** A checkpoint-aware, GPU-efficient implementation completing the full $N=2500$ evaluation on TriviaQA dev, demonstrating production-quality execution for this hypothesis.

2. Related Work

2.0.1. 2.1 CONSISTENCY-BASED UNCERTAINTY QUANTIFICATION

Self-consistency as a reliability signal was established by Wang et al. (2022), who showed that sampling N outputs and selecting the most frequent answer reliably improves reasoning task performance — demonstrating that output diversity correlates with uncertainty. Manakul et al. (2023) operationalized this

as SelfCheckGPT: a black-box hallucination detection method that scores consistency across N samples using NLI-based contradiction detection, BERTScore, or n -gram overlap. The NLI variant achieves NonFact AUC-PR of 92.50 vs. 83.21 for log-probability signals on WikiBio, establishing that consistency-based signals outperform log-prob signals in document generation tasks.

The foundational formalization of consistency-based UQ is Semantic Entropy (SE) (Kuhn et al., 2023), which defines uncertainty as the entropy over semantic equivalence classes: $H(p(C|x)) = -\sum_C p(C|x) \log p(C|x)$, where $p(C|x) = \sum_{s \in C} p(s|x)$ sums token probabilities of all outputs in the same NLI-defined cluster. Kuhn et al. demonstrate that SE outperforms raw token entropy on TriviaQA and BioASQ (AUROC 0.83 on TriviaQA with OPT-30B). The present work directly builds on this formulation ($N=5$, bidirectional DeBERTa-MNLI NLI clustering).

Nguyen et al. (2025) (SNNE) introduced pairwise SE computation to handle multi-sentence answers where the original SE formulation weakens. This directly motivates the present work’s scope restriction to short-answer factual QA (TriviaQA, ≤ 50 token answers), where original SE remains the strongest consistency-based signal. Bouchard and Chauhan (2025) (UQLM) benchmark 50+ UQ methods across 24 model-dataset combinations and find that NLI-based black-box scorers are best in 13/24 AUROC scenarios, providing strong external validation that SE-family signals are competitive.

Gap: None of these works directly measure the Pearson correlation between SE and `min_logprob` on open-weight LLMs. They report standalone AUROC comparisons, not cross-signal statistical independence. The present work closes this gap.

2.0.2. 2.2 TOKEN LOG-PROBABILITY SIGNALS FOR HALLUCINATION DETECTION

Token log-probabilities are the most computationally efficient UQ signal class, requiring only a single forward pass. Malinin and Gales (2021) formalize sequence-level uncertainty measures from log-probabilities. `Min_logprob` has been empirically shown to achieve strong single-pass baseline performance on TriviaQA dev with Llama-3.1-8B (AUROC approximately 0.825; internal pipeline experiment hml).

The relationship between first-token confidence and SE was characterized by Gabriel (2026), who finds $r = 0.54\text{--}0.76$ between first-token probability and seman-

tic agreement in Llama-3.1-8B, TriviaQA — suggesting that for first-token signals, moderate correlation with SE exists. A critical distinction: `min_logprob` is not first-token confidence. It is the minimum over all tokens in greedy decode, capturing arbitrarily late positions. This operational difference drives near-zero Pearson r in the measurements reported here ($|r| = 0.026$).

Dey et al. (2025) (UAF Ensemble) show that fusing UQ signals improves factual accuracy by 8%, further motivating the ensemble direction.

Gap: No prior work directly measures the statistical independence of `min_logprob` and SE on open-weight LLMs under a controlled evaluation protocol. The $|r|=0.026$ measurement reported here fills this gap.

2.0.3. 2.3 LM-AS-A-JUDGE EVALUATION METHODOLOGY

Santilli et al. (2025) show that AUROC rankings for hallucination detection methods are systematically distorted by response length bias — ROUGE-L evaluation is most severely affected. They recommend LM-as-a-judge with cross-model design as the most human-aligned correctness function. Xiong et al. (2023) benchmark consistency-based signals as the most reliable black-box UQ method overall.

The present work implements the cross-model judge recommendation directly: Qwen-2.5-7B-Instruct evaluates Llama-3.1-8B-Instruct outputs on TriviaQA dev. The resulting Spearman $\rho(\text{SE}, \text{judge}) = -0.026$ directly validates this design choice, satisfying the circularity threshold of $|\rho| < 0.40$ with substantial margin.

3. Methodology

3.0.1. 3.1 SIGNAL DEFINITIONS

Semantic Entropy (SE_N5). Following Kuhn et al. (2023), Semantic Entropy is defined as:

$$\text{SE} = H(p(C|x)) = - \sum_{C \in \mathcal{C}} p(C|x) \log p(C|x)$$

where $p(C|x) = \sum_{s \in C} p(s|x)$ aggregates the token-probability mass of all $N=5$ stochastic samples within each NLI-defined semantic cluster C . Clusters are formed by bidirectional NLI entailment using `cross-encoder/nli-deberta-v3-small`. SE operates over $N=5$ stochastic samples (temperature=0.7, top-p=0.95), aggregating uncertainty at the semantic, sentence level. It does not depend on token-level probabilities from the greedy decode path.

Minimum Log-Probability (min_logprob).

$$\text{min_logprob} = \min_{t \in [1, T]} \log p(w_t | w_{< t}, x)$$

where the minimum is taken over all tokens in the greedy decode output of length T . `min_logprob` captures the "weakest link" in the model's token-level confidence chain and is computed from a single greedy decoding pass separate from the $N=5$ stochastic sampling passes used for SE.

Algebraic Distinctness. The near-orthogonality claim follows from: (a) SE aggregates over the semantic clustering distribution of N stochastic outputs; `min_logprob` aggregates as the minimum over a greedy-path token sequence; (b) SE is computed on stochastic samples (temperature > 0); `min_logprob` on the greedy decode (temperature $= 0$). The operations are not mathematically reducible to each other.

3.0.2. 3.2 INDEPENDENCE TEST PROTOCOL

Gate 1 — Pearson Correlation Test. Pre-registered threshold: PASS if $|r| < 0.70$; ABANDON if $|r| > 0.85$ (SE is a reparameterization of `min_logprob`); EXPLORE if $0.70 \leq |r| \leq 0.85$.

Gate 2 — Conditional Logistic Regression (Partial R^2). The following model is fit:

$$\text{logit}(h = 1) \sim \beta_0 + \beta_1 \cdot \text{min_logprob} + \beta_2 \cdot \text{SE} + \beta_3 \cdot L + \beta_4 \cdot (\text{SE} \times \text{min_logprob})$$

Partial $R^2(\text{SE}) = \text{McFadden } R^2(\text{full}) - \text{McFadden } R^2(\text{reduced without SE})$, tested via LRT (2 df). PASS if partial $R^2 \geq 0.02$ AND LRT $p < 0.05$.

3.0.3. 3.3 CIRCULARITY CONTROL DESIGN

A cross-model LM-judge (Qwen-2.5-7B-Instruct evaluating Llama-3.1-8B-Instruct outputs) prevents shared model-specific NLI biases from creating circular agreement between SE and judge labels. Circularity diagnostic: Spearman $\rho(\text{SE}, \text{judge_correctness})$; threshold: $|\rho| < 0.40$.

3.0.4. 3.4 IMPLEMENTATION

Six modules compose the pipeline: `config.py` (hyperparameters), `generate.py` (checkpoint-aware stochastic and greedy generation), `compute_signals.py` (SE_N5 and `min_logprob`), `judge.py` (Qwen-2.5-7B LM-judge), `stats_analysis.py` (Pearson r , conditional LR, partial R^2 , gate evaluation), `visualize.py` (four figures). The checkpoint-aware pipeline saves sig-

nals to `results/signals.pkl`. GPU memory is managed explicitly between generator and judge loads.

4. Experimental Setup

Research Questions:

- **RQ1:** Are SE_N5 and `min_logprob` statistically independent uncertainty signals?
- **RQ2:** Does SE contribute predictive signal for correctness beyond `min_logprob`?
- **RQ3:** Does the cross-model judge design produce low circularity?

4.0.1. 4.1 DATASET

TriviaQA dev (rc.nocontext). The closed-book variant of TriviaQA dev (mandarjoshi/trivia_qa, rc.nocontext, validation split), first $N=2500$ prompts (seed=42). TriviaQA is a challenging factual QA benchmark where hallucination is frequent, providing sufficient error diversity for UQ evaluation. The correctness rate as assessed by LM-judge is 34.8% (870/2500 prompts). The no-context variant isolates parametric knowledge from retrieval artifacts, matching the prior baseline configuration.

The $N=2500$ evaluation constitutes the primary experimental record. It was preceded by a proof-of-concept run at $N=300$ that validated pipeline correctness, yielding Pearson $|r|=0.049$, partial $R^2 = 0.0101$, LRT $p=0.1504$, Spearman $\rho(\text{SE}, \text{correctness})=-0.081$, and correctness rate 34.3% (103/300).

Property	Value
Dataset	TriviaQA dev, rc.nocontext
N prompts	2500, seed=42
Task type	Closed-book short-answer factual QA
Answer format	Free-form, ≤ 50 tokens
Correctness rate (LM-judge)	34.8% (870/2500)

4.0.2. 4.2 MODELS

Generator: Llama-3.1-8B-Instruct (bfloat16, device_map="auto"). $N=5$ stochastic samples per prompt (temperature=0.7, top_p=0.95, max_new_tokens=50). One greedy decode per prompt for `min_logprob` extraction.

NLI Backbone: cross-encoder/nli-deberta-v3-small for bidirectional NLI clustering (batch size 32).

LM-Judge: Qwen-2.5-7B-Instruct (cross-model; Llama outputs evaluated by a different model family; batch size 16).

4.0.3. 4.3 BASELINES

min_logprob (standalone): The minimum token log-probability from a single greedy decode, achieving AUROC approximately 0.825 on TriviaQA dev (internal pipeline experiment h-m1, not published externally). Reference signal for independence test (RQ1) and conditional LR (RQ2).

SE_N5 (standalone): Semantic Entropy from N=5 stochastic samples per Kuhn et al. (2023). The alternative standalone signal; focus of independence test against min_logprob. Standalone AUROC for SE_N5 is not the focus of this work; this experiment tests independence preconditions, not standalone ranking.

Note: Ensemble AUROC is not evaluated in this work. The experiment tests independence preconditions. Given the null result for partial R^2 , linear ensemble AUROC improvement is not predicted; follow-up work investigates nonlinear combinations.

4.0.4. 4.4 EVALUATION METRICS AND HYPERPARAMETERS

5. Results

5.0.1. 5.1 MAIN RESULT: SE_N5 AND MIN_LOGPROB ARE NEAR-ORTHOGONAL (RQ1)

Pearson $|r|(\text{SE_N5, min_logprob}) = 0.026$ at N=2500 on TriviaQA dev with Llama-3.1-8B-Instruct. This value is far below both the independence threshold (0.70) and the reparameterization boundary (0.85), meeting Gate 1 with substantial margin.

Figure 1 (h-e1/figures/scatter_se_vs_minlogprob.png) visualizes this independence: 2500 data points show no discernible linear or monotone pattern between SE and min_logprob values, with correctness labels scattered throughout the joint space. The ABANDON threshold was not approached, ruling out SE as a monotone reparameterization of log-probability.

Figure 2 (h-e1/figures/correlation_heatmap.png) shows the full Pearson correlation matrix between SE_N5, min_logprob, response length (L), and LM-judge correctness. The near-zero cross-correlation between SE and min_logprob ($|r|=0.026$) stands in contrast to the moderate negative correlations that both signals exhibit individually with correctness — each carries information about correctness, but through independent mechanisms.

The mechanism reality checks confirm both signals are well-behaved: SE variance = 0.100 (well above non-degeneracy threshold), fraction_degenerate = 0 (no

prompts where all 5 stochastic samples collapse to the same semantic cluster), and min_logprob mean = -2.435 (all values are valid negative log-probabilities).

5.0.2. 5.2 CONDITIONAL INDEPENDENCE ANALYSIS (RQ2)

Figure 3 (h-e1/figures/gate_metrics.png) summarizes the gate evaluation: the Pearson $|r|=0.026$ bar clears the 0.70 threshold with wide margin, while the partial $R^2 = 0.0005$ bar falls below the 0.02 threshold (LRT $\chi^2 = 1.632$, $df=2$, $p=0.442$, non-significant). Figure 4 (h-e1/figures/lr_coefficients.png) shows the conditional logistic regression coefficients with 95% CI. The SE coefficient is positive (correct direction: higher SE predicts lower correctness), but the confidence interval crosses zero at N=2500 — consistent with a powered test yielding a genuine null result.

The partial $R^2 = 0.0005$ at N=2500 does not meet the pre-registered ≥ 0.02 threshold. The LRT provides no evidence that SE adds conditional predictive power beyond min_logprob in a linear setting.

Gate outcome: The pre-registered pipeline gate classified this result as EXPLORE_N10 (gate_pass: false, reason: "abs(r)=0.026 or partial_r2=0.0005 marginal — retry N=10"). The gate did not pass. The EXPLORE_N10 decision indicates the pipeline recommends extending to N=10 stochastic samples per prompt. A standard power analysis for logistic regression at N=2500 with binary outcome prevalence of 34.8% indicates that N=2500 substantially exceeds the sample size required to detect partial $R^2 \geq 0.02$ at $\alpha = 0.05$ with >0.80 power (estimated power >0.99 at the pre-registered effect size). The gate's EXPLORE_N10 flag reflects the pipeline's conservative marginal-uncertainty criterion applied to both metrics jointly, not a determination that the statistical test was underpowered for the pre-registered threshold. These two characterizations are not contradictory: the gate reflects pipeline confidence policy; the power characterization reflects adequacy of N=2500 to detect the pre-registered effect size.

5.0.3. 5.3 CIRCULARITY CONTROL VALIDATION (RQ3)

Spearman $\rho(\text{SE, LM-judge correctness}) = -0.026$, well within the $|\rho| < 0.40$ circularity threshold. The negative sign is directionally consistent: higher SE (more semantic uncertainty) correlates weakly with lower correctness. The small magnitude confirms that the cross-model judge design prevents circular agreement between the NLI-based SE signal and the judge.

Metric	Purpose	Pre-registered Threshold	
Pearson	r	(SE_N5, min_logprob)	Primary independence metric
Partial R^2 (SE) in conditional LR	Conditional predictive contribution	≥ 0.02 , LRT $p < 0.05$	< 0.70 (PASS); > 0.85 (ABANDON)
Spearman ρ (SE, judge_correctness)	Circularity diagnostic		ρ
SE variance	Non-degeneracy check	> 0.01	< 0.40
Fraction degenerate	Non-degeneracy check	$= 0$	

Hyperparameter	Value
N stochastic samples	5
Temperature (stochastic)	0.7
Top-p	0.95
Max new tokens	50
NLI batch size	32
Judge batch size	16
Pearson r threshold	0.70
Partial R^2 threshold	0.02
Circularity threshold (ρ)	0.40
ABANDON threshold	0.85

Note: the Spearman $|\rho|=0.026$ and the Pearson $|r|=0.026$ reported in Section 5.1 share the same rounded magnitude; these are numerically distinct statistics computed on different variable pairs (SE vs. min_logprob for Pearson r; SE vs. judge labels for Spearman ρ), using algebraically different correlation measures. The coincidence is real — both reflect a near-zero correlation structure in this dataset — and is not a copy-paste artifact.

The correctness rate of 34.8% provides meaningful error headroom for UQ discrimination.

5.0.4. 5.4 ANALYSIS: WHY $|r|=0.026$ IS SUBSTANTIALLY LOWER THAN PRIOR WORK

Gabriel (2026) reports $r = 0.54\text{--}0.76$ between **first-token** confidence and semantic agreement on Llama-3.1-8B, TriviaQA. The min_logprob result ($|r|=0.026$) is approximately $20\times$ lower. This contrast is mechanistically explained: min_logprob is the minimum over the full greedy decode sequence (capturing late-position tokens where factual information is concentrated), not the first token. Late-position tokens can be highly uncertain even when the first token is confident, breaking the correlation that exists for first-token signals.

5.0.5. 5.5 SUMMARY

6. Discussion

6.0.1. 6.1 KEY FINDINGS AND THEIR IMPLICATIONS

SE and min_logprob measure genuinely distinct uncertainty dimensions. The Pearson $|r|=0.026$ is not merely a marginal pass of the 0.70 threshold — it indicates near-orthogonality by any standard metric for variable collinearity. This gap from the prior first-token measurement ($r=0.54\text{--}0.76$) is mechanistically explained: min_logprob captures late-sequence token uncertainty while first-token confidence captures only the initial prediction. The operational separation between SE (stochastic sampling, semantic aggregation) and min_logprob (greedy decode, token-level minimum) manifests as near-zero empirical correlation.

This finding establishes a necessary precondition for ensemble benefit: it rules out the scenario where combining SE and min_logprob provides no benefit because one is a redundant reparameterization of the other. However, necessity is not sufficiency — and the partial R^2 null result (0.0005 at $N=2500$) shows that independence does not automatically yield linear ensemble gain. The signals are empirically near-orthogonal yet carry redundant correctness-predictive information in the linear logistic regression setting.

Cross-model judge design suppresses circularity effectively. The $\rho(\text{SE}, \text{judge}) = -0.026$ result validates the evaluation protocol: using Qwen-2.5-7B to evaluate Llama-3.1-8B outputs prevents shared model-specific NLI biases from creating circular agreement. This design choice, recommended by Santilli et al. (2025), is empirically validated in this setting.

The partial R^2 null result: power characterization and gate disclosure. The partial $R^2 = 0.0005$ at $N=2500$ (LRT $p=0.442$) was classified by the pre-registered pipeline gate as EXPLORE_N10 (gate_pass: false). This outcome is disclosed transparently. At $N=2500$, estimated statistical power exceeds 0.99 for detecting the pre-registered effect size of partial $R^2 \geq 0.02$ at $\alpha = 0.05$. The gate’s EXPLORE_N10 flag reflects the pipeline’s conservative joint marginal-uncertainty criterion, not a power inadequacy at the pre-registered threshold. We therefore characterize the

Metric	Value	Threshold	Status
Pearson	r	(SE_N5, min_logprob)	**0.026**
ABANDON threshold	—	> 0.85	< 0.70 **PASS**
SE variance	0.100	> 0.01	NOT triggered
Fraction degenerate	0.000	= 0	PASS (non-degenerate)
min_logprob mean	−2.435	< 0	PASS

LR Term	Direction	Significance	p-value
min_logprob	positive	significant (CI above zero)	< 0.05
SE_N5	positive	not significant (CI crosses zero)	> 0.05
Partial $R^2(SE)$	—	**0.0005** (target ≥ 0.02)	p = 0.442

partial $R^2 = 0.0005$ finding as a powered null for the pre-registered effect size: evidence of absence of a linear conditional effect at the pre-registered threshold on this benchmark, rather than absence of evidence due to underpowering. The dissociation between near-zero Pearson correlation ($|r|=0.026$) and near-zero conditional contribution (partial $R^2 = 0.0005$) indicates that, while the signals are computed differently, they may capture overlapping correctness-predictive information in the linear logistic regression setting.

6.0.2. 6.2 LIMITATIONS

Ensemble AUROC not measured; linear null bounds linear gain. Ensemble AUROC is not evaluated in this work. The partial R^2 null at $N=2500$ (gate=EXPLORE_N10) indicates linear logistic regression ensembles will not achieve the pre-registered $\Delta AUROC$ target of ≥ 0.025 . Ensemble AUROC on TriviaQA dev and cross-model transfer to Qwen-2.5-7B on TruthfulQA are neither confirmed nor refuted; these are primary targets for follow-up work (h-e1-v2), which will also extend to $N=10$ stochastic samples per prompt as the gate recommends.

Single seed and no bootstrapped confidence intervals. $N=2500$ with a fixed seed (42) provides no estimate of result variance. The Pearson $|r|=0.026$ and Spearman $\rho = -0.026$ are point estimates. Bootstrapped 95% CIs at $N=2500$ are planned for the h-e1-v2 analysis alongside nonlinear ensemble evaluation.

Scope: short-answer factual QA with 7–8B open-weight models. Results are validated on TriviaQA dev (≤ 50 token answers) with Llama-3.1-8B-Instruct. Generalization to long-form generation is explicitly out of scope — Nguyen et al. (2025) document SE’s weakness for multi-sentence answers.

Unverified references. The Gabriel (2026) paper (arXiv:2605.05166) and Santilli et al. (2025) (arXiv:2504.13677) were not found in Semantic Scholar

at the time of writing; they are cited as preprints. The comparison finding from Gabriel (2026) ($r=0.54$ – 0.76 for first-token confidence) is used only to motivate the contrast with min_logprob, not as a load-bearing empirical claim.

6.0.3. 6.3 BROADER IMPACT

This work contributes to the safety and reliability of LLM deployments by establishing the empirical independence of two lightweight UQ signals — one requiring only greedy decode access (min_logprob) and one requiring only generation API access (SE_N5). The replicable evaluation methodology (cross-model LM-judge, circularity diagnostic, mechanism reality checks) provides a template for future UQ signal characterization studies. The pipeline uses publicly available open-weight models, ensuring reproducibility without proprietary API dependence.

7. Conclusion

This paper directly measured the statistical independence of SE_N5 and min_logprob as uncertainty signals for hallucination detection in open-weight LLMs on short-answer factual QA. On TriviaQA dev with Llama-3.1-8B-Instruct across $N=2500$ prompts, the Pearson correlation between these signals is $|r| = 0.026$ — near-orthogonal, far below both the pre-registered independence threshold (0.70) and the reparameterization boundary (0.85). This result is not a marginal pass; it indicates that SE, which aggregates semantic entropy over $N=5$ stochastic samples, and min_logprob, which extracts the minimum token confidence from a single greedy decode, operate as genuinely distinct uncertainty mechanisms.

The main contributions are: (1) Pearson $|r|=0.026$ confirming near-orthogonality at $N=2500$; (2) partial $R^2 = 0.0005$ (LRT $p=0.442$, LRT $\chi^2 = 1.632$, $df=2$) showing that independence does not yield linear conditional predictive utility on this benchmark — a powered null result for the pre-registered effect size, with the gate outcome (EXPLORE_N10) disclosed transparently; (3) Spearman $\rho(SE, \text{judge})=-0.026$ validating the cross-model LM-judge design as a low-

RQ	Question	Finding	Status
RQ1	Are SE and min_logprob independent?	Pearson	r = 0.026 — near-orthogonal **CONFIRMED**
RQ2	Does SE add conditional predictive power?	Partial $R^2 = 0.0005$, LRT $p=0.442$ at $N=2500$ — null result (gate=EXPLORE_N10)	**NOT CONFIRMED**
RQ3	Is cross-model judge low-circularity?	Spearman $\rho = -0.026$	**CONFIRMED**

circularity evaluation protocol; and (4) a fully reproducible, checkpoint-aware pipeline that completed the N=2500 evaluation.

Future work includes: nonlinear ensemble evaluation (h-e1-v2); extending to N=10 stochastic samples per prompt per the gate recommendation; cross-model transfer characterization (Llama-calibrated ensemble on Qwen-2.5-7B TruthfulQA); orthogonality zone analysis; bootstrapped CIs on primary metrics; and seed-stability verification. The findings motivate combining these signals through nonlinear methods rather than linear logistic regression. Careful statistical characterization of signal independence — not just standalone AUROC comparison — is the appropriate foundation for principled ensemble design in LLM hallucination detection.

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